INCH-POUND
MIL-DTL-18015C(SH)
8 May 2019
SUPERSEDING
MIL-G-18015B(SH)
2 May 1994

DETAIL SPECIFICATION

GRATINGS, METAL, OTHER THAN BAR TYPE (SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum and steel floor gratings other than bar types for shipboard use.

1.2 Classification. Gratings are of the following types and classes, as specified (see 6.2).

1.2.1 Types.

a. Type I – Steel, heavy
   (1) Class 1 – Galvanized
   (2) Class 2 – Ungalvanized
   (3) Class 3 – Corrosion resistant
b. Type II – Aluminum, heavy
c. Type III – Steel, light
   (1) Class 1 – Galvanized
   (2) Class 2 – Ungalvanized
   (3) Class 3 – Corrosion resistant
d. Type IV – Aluminum, light

1.3 Part or identifying number (PIN). PINs to be used for gratings acquired to this specification are created as follows:

M	**18015**	**-**	**X**	**-**	**X**
Prefix for Military Specification	Specification Number		Type (I, II, III, or IV)		Class (for Types I and III) (1, 2, or 3)
Examples: M18015-I-2, M18015-IV					

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5670

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:53Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection of Attributes

(Copies of this document are available online at http://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM A153/A153M	-	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B211	-	Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire
ASTM B221	-	Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B241/B241M	-	Standard Specification for Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube
ASTM F1267	-	Standard Specification for Metal, Expanded, Steel

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

3.2.1 Steel. Types I and III gratings shall be made of either carbon steel (rolled or drawn), high tensile alloy steel (rolled or drawn), or CRES 304 (rolled or drawn) as specified in ASTM B241/B241M, ASTM B221, ASTM B211, or ASTM B209.

3.2.2 Aluminum. Types II and IV gratings shall be made of material meeting the chemical requirements of ASTM B241/B241M, ASTM B221, ASTM B211, or ASTM B209. When the material is not specified, the manufacturer may furnish any of these materials (see 3.2 and 6.2).

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:53Z
Check the source to verify that this is the current version before use.

3.3 Construction. The grating panel sizes shall be as specified (see 6.2). Where nonrectangular shapes or cutouts in rectangular panels are required, they shall be as shown on the drawings as specified (see 6.2). The edges of nonrectangular shapes and cutouts where more than four bearing bars or channels are included, or where the expanded metal is cut, shall be finished with banding bars of the same depth as the bearing bars. The banding bars when required shall be welded to the bearing bars or channels of the grating at least every fourth bearing bar or channel and in every corner. Intervening bearing bars or channels shall be tack welded. The ends of 6000 series aluminum grating panels shall be finished or banded by means of spacers or bridges to preclude welding. The 5000 series aluminum may be welded. The weight of gratings (pounds per square foot) shall not exceed the limits of table I.

3.3.1 Clamps and fastenings. Clamps or other fastenings shall be provided when specified (see 6.2). Such clamps or fastenings shall be: of the same material and finished in the same manner as the gratings; shall not extend above the surface of the gratings; and shall be of such construction as to provide means for holding and securing the gratings to the supporting structure. Clamps or fastenings shall be designed for simplicity of installation and for removal and replacement in service.

3.3.1.1 Thickness and spacing. The thickness and spacing of all bearing bars or channels and the depth, thickness, and spacing of all cross or rectilinear bars or expanded metal and frame shall be capable of supporting the loads specified in table I without deflection due to bending, and without lateral displacement, warping, or distortion of any of the parts greater than 0.2 inch for steel, or 0.5 inch for aluminum, and without permanent set greater than 0.005 inch (see 4.6.2).

3.3.2 Strength and load. Gratings shall be of such construction (except cross-bar) as to meet the strength and load requirements of this specification (see 3.6). They shall be of expanded metal, extruded and welded sections, or of other construction (except cross-bar). Unless otherwise specified (see 6.2), gratings having serrated surface members are permitted. Supports for gratings, if required, shall be of the dimensions specified (see 6.2).

3.3.2.1 Steel expanded metal. Steel expanded metal gratings shall meet the requirements of ASTM F1267 and 3.6 of this specification, unless otherwise specified (see 6.2).

3.3.3 Openings. Unless otherwise specified (see 6.2), the dimensions of the openings for expanded-metal gratings shall not exceed 15⁄16 by 3 7⁄16 inches. The dimension of the openings for other types of gratings shall not exceed 1 inch by 4 inches.

3.4 Tolerances – general. Tolerances shall be in accordance with the following:

a. Length of bearing bars or channels – ±1⁄16 inch in 6 feet.

b. Depth of bearing bars or channels – ±1⁄32 inch (when measured from bottom of serrations when present).

c. Width of grating panels – ±1⁄8 inch.

d. Openings – ±10 percent.

A tolerance of plus 10 percent is permitted on the maximum weight of any individual grating when banding bars or fasteners or both form a part of the grating.

3.5 Upper surface. The upper surface of gratings shall be free from sharp corners, burrs, and projections (which may tear or catch cloth or paper bags of provisions dragged over the gratings).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:53Z
Check the source to verify that this is the current version before use.

3.6 Load. The load requirements of the gratings shall be as shown in table I.

TABLE I. Load requirements.

	Steel			Aluminum		
Type	**Design load per square foot (pounds)**	**Test load 1/ (pounds)**	**Maximum weight per square foot 2/ (pounds)**	**Design load per square foot (pounds)**	**Test load 1/ (pounds)**	**Maximum weight per square foot 2/ (pounds)**
I	300	900	5.00	---	---	---
II	---	---	---	300	900	5.00
III	175	525	4.40	---	---	---
IV	---	---	---	150	450	3.70

NOTES:

1/ Concentrated midway between support 2 feet long, on a 3-foot span (see 4.6).

2/ Finished, unbanded, including paint, bearing bars, or channels, and on types I and III, zinc-coating (galvanized).

3.7 Zinc-coating (galvanized). Unless otherwise specified (see 6.2), steel gratings shall be completely and evenly zinc-coated (galvanized) by the hot-dip process in accordance with ASTM A153/A153M after fabrication. The coating shall not flake off when tested as specified (see 4.6.3.1).

3.8 Workmanship. Gratings shall be fabricated accurately and shall be free from warps, twists, and other defects affecting appearance or serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on gratings when a first article sample is required (see 3.1). This inspection shall include the examination of 4.5 and the test of 4.6.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5.

4.4 Sampling.

4.4.1 Lot. For purposes of sampling, a lot shall consist of all gratings of the same type, class, and size offered for delivery at the same time.

4.4.2 Sampling for examination. A random sample of grates shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and Acceptance Quality Limit (AQL) as specified (see 6.2).

4.5 Examination. Each sample (see 4.4.2) shall be examined to determine conformance with material requirements (see 3.1), construction and weight (see 3.2), dimensional tolerance (see 3.3), zinc coating (see 3.7), and workmanship (see 3.4 and 3.8).

4.6 Load test. Each sample item shall be subjected to a load test. Each item shall be supported by two supports with bearing width not to exceed 1 inch. The supports shall be placed 3 feet apart. The gratings shall be tested with the load specified for the type and class in accordance with table I, suspended from a bar placed midway between the supports. Deflection and permanent set shall be determined by gages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:53Z
Check the source to verify that this is the current version before use.

4.6.1 Deflection criteria. The maximum deflection measured at the center of the grating shall not exceed 0.2 inch. There shall be no visible lateral displacement, warping, or distortion (see 3.3.1.1).

4.6.2 Permanent set criteria. The maximum permanent set measured at the center of the grating shall not exceed 0.005 inch (see 3.3.1.1).

4.6.3 Zinc-coating.

4.6.3.1 Adhesion of coating. The sample gratings selected in accordance with 4.4.2 shall be struck a hard glancing blow with a straight peen or scaling hammer having all edges ground to a radius of not less than 3/64 inch and weighing approximately 1 pound. The coating shall not flake off.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Gratings manufactured to this specification are intended for use in a maritime environment with exposure to salt and heavy loads. Gratings are expected to function for the life of a naval ship and withstand shock loadings associated with a military environment. When bar type gratings are required, MIL-DTL-18014 should be used. When the intended use does not restrict the type of grating, either MIL-DTL-18014 or this specification should be used. Current stock of gratings may be used until depleted; future requisitions for gratings should refer to the current version of this specification. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of gratings required (see 1.2).
c. Whether first article is required (see 3.1).
d. Specification requirements for types II and IV gratings (see 3.2.2).
e. Size of grating panels and drawings required (see 3.3).
f. Whether clamps or fastenings are required (see 3.3.1).
g. When gratings having serrated surface members are not permitted (see 3.3.2).
h. Supports for gratings, if required (see 3.3.2).
i. Steel expanded metal gratings requirements (see 3.3.2.1).
j. Dimensions of openings when larger than the maximum specified (see 3.3.3).
k. Special instructions for zinc-coating (see 3.7).
l. AQL 1.0 (see 4.4.2).
m. Packaging requirements (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:53Z
Check the source to verify that this is the current version before use.

6.3 Typical grating figure. Figure 1 represents an arrangement of banding, bearing, and cross bars for reference purposes. Expanded metal grating is not shown.

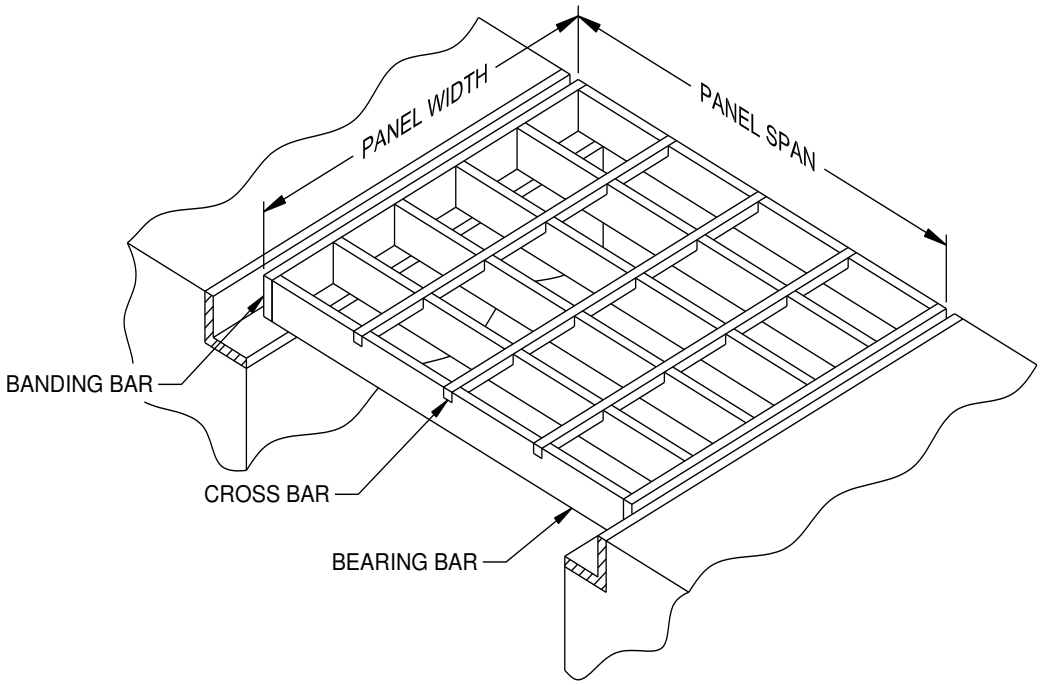


FIGURE 1. Typical grating.

6.4 Subject term (keyword) listing.

Aluminum

Corrosion resistant steel

Floor gratings

Galvanized steel

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5670-2019-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:53Z
Check the source to verify that this is the current version before use.